

**PROJECT
ON
“E-Commerce Sales Data Analysis”**

**COMPLETED AT
LYALLPUR KHALSA COLLEGE OF ENGINEERING**



Submitted for the partial fulfillment for the award degree of
Bachelor of Technology
IN
(Computer Science & Engineering)

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CERTIFICATE

DECLARATION

I, Khushi Kumari, Roll No. 2443083, hereby declare that the project report titled “E-Commerce Sales Data Analysis” is an authentic record of my own work carried out as part of the 6-Week Data Mining (DM) Training under the guidance of my faculty mentor at Lyallpur Khalsa College Technical Campus, Jalandhar.

The analysis, findings, and interpretations presented in this report are based on the dataset provided/collected for this project and have been prepared using Python and its data-analysis libraries. This work has not been submitted elsewhere for the award of any degree, diploma, or other similar title.

Date: _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty mentor and the Department of Computer Applications / Computer Science, Lyallpur Khalsa College Technical Campus, Jalandhar, for their invaluable guidance, encouragement, and support throughout the duration of this project.

I am thankful to the college for providing the necessary resources, infrastructure, and a conducive learning environment that made it possible for me to complete this 6-Week Data Mining Training and the accompanying project report on “E-Commerce Sales Data Analysis.”

I would also like to thank my family and friends for their constant motivation and support. Finally, I acknowledge the open-source Python community, whose tools and libraries — including pandas, NumPy, and Matplotlib — formed the technical backbone of this analysis.

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ABSTRACT

Electronic commerce (e-commerce) has fundamentally transformed the way consumers purchase goods, and every transaction generates data that can be mined for strategic insight. This project presents a comprehensive analysis of an e-commerce sales dataset comprising 1,000 transaction records spanning January 2023 to January 2024, covering customer demographics, product categories, regional distribution, pricing, and order fulfilment status.

The study follows a structured data-mining approach: data collection, data cleaning and pre-processing, exploratory data analysis (EDA), visualization, and interpretation. Python, along with the pandas, NumPy, and Matplotlib libraries, was used to clean inconsistent records, handle missing values, and generate seventeen distinct visualizations covering revenue trends, category and regional performance, customer demographics, order fulfilment, and correlation analysis.

Key findings reveal that the Electronics category — led by Laptops — dominates revenue generation, contributing close to nine-tenths of total sales, while the customer base is fairly evenly split across gender and geographic region. The analysis further uncovers a notable return rate that warrants managerial attention, along with seasonal peaks in sales during January and November. Based on these findings, the report concludes with actionable recommendations for inventory planning, customer targeting, and return-rate reduction, along with directions for future work such as predictive modelling and customer segmentation.

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Chapter 1: Introduction

1.1 Background

Electronic commerce, popularly known as e-commerce, refers to the buying and selling of goods and services over the internet. Over the last decade, e-commerce has grown from a niche convenience into one of the primary channels of retail commerce worldwide, driven by increasing internet penetration, smartphone adoption, and changing consumer preferences. Platforms such as Amazon, Flipkart, and Myntra process millions of transactions every day, and each transaction leaves behind a digital trail — the order value, product category, customer demographics, shipping details, and much more.

This continuous stream of transactional data represents a vast and largely untapped resource. When properly collected, cleaned, and analyzed, it can reveal patterns in customer behaviour, product performance, seasonal demand, and operational bottlenecks that are invisible to the naked eye. This is where the discipline of data mining and data analytics becomes indispensable — transforming raw, often messy, transactional records into actionable business intelligence.

1.2 Motivation for the Study

The motivation behind this project stems from the growing importance of data-driven decision-making in the retail and e-commerce sector. Businesses that can accurately interpret their sales data are better positioned to optimize inventory, design targeted marketing campaigns, forecast demand, and improve customer satisfaction. As part of the 6-Week Data Mining (DM) Training, this project was undertaken to apply core data-mining concepts — data cleaning, exploratory data analysis, visualization, and interpretation — to a real-world-style e-commerce sales dataset, thereby bridging the gap between classroom theory and practical application.

1.3 About the Dataset

The dataset used in this study contains 1,000 individual e-commerce transaction records collected over a twelve-month period (January 2023 to January 2024). Each record captures information about the customer (ID, gender, age, region), the product purchased (name, category, unit price, quantity), the resulting order value, the shipping fee charged, the shipping/fulfilment status, and the date of the order. This combination of demographic, product, and operational attributes makes the dataset well suited for a multi-dimensional sales analysis.

1.4 Why Data Analysis Matters in E-Commerce

- It helps identify best-selling and under-performing products, guiding inventory and procurement decisions.
- It reveals which customer segments (by age, gender, or region) generate the most revenue, enabling targeted marketing.
- It exposes seasonal or monthly demand patterns that support better forecasting and stock planning.
- It highlights operational issues such as high return rates or delayed shipments that affect profitability and customer trust.
- It supports evidence-based strategy formulation rather than intuition-driven decisions.

1.5 Report Organization

This report is organized into thirteen chapters. Chapter 2 outlines the objectives of the study. Chapter 3 defines the scope and limitations. Chapter 4 presents a brief literature review of related work in sales-data analytics. Chapter 5 describes the tools and technologies used. Chapter 6 details the research methodology. Chapter 7 describes the dataset in depth. Chapter 8 documents the data-cleaning process. Chapter 9 forms the core of the report, presenting the exploratory data analysis with seventeen visualizations and their interpretations. Chapters 10 through 13 present the key findings, recommendations, conclusion, and future scope respectively, followed by references and an appendix containing the Python source code used for the analysis.

Chapter 2: Objectives of the Study

The primary aim of this project is to analyze e-commerce sales data to extract meaningful, actionable business insights. The specific objectives are:

1. To collect, clean, and pre-process the raw e-commerce sales dataset, addressing missing values, inconsistent entries, and outliers.
2. To perform exploratory data analysis (EDA) to understand the distribution and relationships among key variables such as revenue, category, region, and customer demographics.
3. To identify top-performing products and categories in terms of revenue and order volume.
4. To analyze regional and demographic (age and gender) sales patterns to understand the customer base.
5. To study monthly and quarterly sales trends and identify seasonality in demand.
6. To examine order fulfilment and shipping status, including the analysis of return rates by category.
7. To visualize all findings using clear, interpretable charts created with Python's Matplotlib library.
8. To derive data-driven recommendations that could help an e-commerce business improve revenue, customer satisfaction, and operational efficiency.

Chapter 3: Scope and Limitations

3.1 Scope

This study focuses exclusively on descriptive and exploratory analysis of the provided e-commerce sales dataset. It covers sales performance across products, categories, regions, and time periods, along with customer demographic analysis and order-fulfilment analysis. The analysis is performed entirely using Python (pandas, NumPy, and Matplotlib) within a reproducible, script-based workflow, and all findings are grounded strictly in the 1,000 transaction records provided.

3.2 Limitations

- The dataset is limited to 1,000 records and 292 unique customers, which, while sufficient for illustrative analysis, is smaller than a typical production e-commerce database.
- The dataset contains only seven distinct products across three categories (Electronics, Accessories, and Wearables), which limits the diversity of category-level insights.
- Certain fields — Region, Age, and Shipping Status — contained missing values that had to be imputed or labelled as “Unknown,” which introduces a small degree of estimation into the affected sub-analyses.
- The analysis is descriptive and exploratory in nature; predictive modelling (e.g., demand forecasting or churn prediction) is identified as future work rather than being part of the current scope.
- External factors such as marketing spend, competitor pricing, and macroeconomic conditions are not available in the dataset and are therefore not part of this analysis.

Chapter 4: Literature Review

Data mining in the retail and e-commerce domain has been an active area of research and industry practice for over two decades. Early work in market-basket analysis focused on association-rule mining to understand which products are frequently purchased together, forming the basis for recommendation engines used by major online retailers today.

With the growth of big data technologies, the focus of e-commerce analytics has expanded from simple descriptive reporting to more sophisticated techniques, including customer segmentation using clustering algorithms (such as K-Means), churn prediction using classification models, and demand forecasting using time-series methods. Customer Relationship Management (CRM) systems increasingly rely on Recency-Frequency-Monetary (RFM) analysis to score and segment customers based on their purchase behaviour.

Exploratory Data Analysis (EDA), the core technique applied in this project, remains foundational to any data-mining pipeline. As emphasized in the broader data-science literature, a thorough EDA phase — involving data cleaning, univariate and bivariate analysis, and visualization — is essential before any predictive modelling is attempted, since it surfaces data-quality issues and hidden patterns that would otherwise bias later models.

In the specific context of e-commerce sales analysis, prior industry studies consistently highlight a few recurring themes that this project also investigates: the concentration of revenue in a small number of high-value product categories (often following a Pareto-like 80/20 distribution), the influence of seasonality on sales volume, regional variation in purchasing behaviour, and the operational impact of returns and cancelled shipments on overall profitability. These themes form the analytical backbone of the present study and are examined in detail in Chapter 9.

Chapter 5: Tools and Technologies Used

This project was implemented entirely using the Python programming language, owing to its rich ecosystem of open-source libraries for data manipulation, analysis, and visualization. The following tools and libraries were used:

5.1 Python

Python was chosen as the primary programming language for this project because of its readability, extensive scientific-computing ecosystem, and widespread adoption in the data-science and data-mining community.

5.2 Pandas

Pandas is a high-performance Python library used for data manipulation and analysis. In this project, pandas was used to load the raw CSV dataset, inspect data types, detect and handle missing values and inconsistent entries, engineer new features (such as Age Group, Month, and Quarter), and perform groupings and aggregations for statistical summaries.

5.3 NumPy

NumPy provides efficient numerical computing capabilities and underlies much of pandas' internal functionality. It was used for numerical operations such as computing means, standard deviations, and correlation coefficients.

5.4 Matplotlib

Matplotlib is a widely used Python plotting library. All seventeen charts presented in Chapter 9 — including bar charts, line charts, pie charts, histograms, and heatmaps — were generated using Matplotlib, customized with a consistent colour palette for visual clarity and professional presentation.

5.5 Microsoft Word / Report Generation

The final project report was compiled and formatted as a Microsoft Word (.docx) document, incorporating the generated charts, tables, and structured written analysis in accordance with standard academic project-report conventions.

Tool / Library	Purpose
Python 3	Core programming language for the entire analysis
Pandas	Data loading, cleaning, transformation, and aggregation
NumPy	Numerical computation and statistical operations
Matplotlib	Data visualization (bar, line, pie, histogram, heatmap charts)
CSV	Source data format for the raw e-commerce dataset
Microsoft Word	Final report compilation and formatting

Chapter 6: Research Methodology

This project follows a standard data-mining and exploratory-analysis workflow, commonly referred to in industry as a simplified CRISP-DM (Cross-Industry Standard Process for Data Mining) approach. The methodology is summarized in the following stages:

6.1 Data Collection

The raw dataset, provided in CSV format, was collected as the starting point of the analysis. It comprises 1,000 rows and 12 columns capturing customer, product, and order-level information.

6.2 Data Understanding

Before any cleaning was performed, the dataset was inspected to understand its structure — column names, data types, value ranges, and the presence of missing or inconsistent values. This step is critical to forming an accurate data-cleaning plan.

6.3 Data Cleaning and Pre-processing

Missing values, duplicate records, and inconsistent numerical entries were identified and treated using appropriate strategies (detailed in Chapter 8), ensuring the dataset was analysis-ready.

6.4 Exploratory Data Analysis (EDA)

Univariate analysis (examining single variables such as revenue or age distribution) and bivariate/multivariate analysis (examining relationships such as revenue by category and region, or gender by category) were both performed to uncover patterns.

6.5 Data Visualization

Each analytical question was paired with an appropriate chart type — bar charts for categorical comparisons, line charts for trends over time, pie charts for proportional breakdowns, a histogram for distribution analysis, and heatmaps for correlation and cross-tabulation analysis.

6.6 Interpretation and Reporting

Finally, the visualized results were interpreted in business terms, and the key findings were consolidated into actionable recommendations, forming the concluding chapters of this report.

9. Data Collection (raw CSV)
10. Data Understanding (structure, types, quality check)
11. Data Cleaning & Pre-processing
12. Exploratory Data Analysis
13. Data Visualization
14. Interpretation & Recommendations

Chapter 7: Data Description

The dataset used for this analysis, “E_commerce_sales_data.csv,” contains 1,000 records and 12 attributes. Table 1 describes each attribute in detail.

Column Name	Description	Data Type
Customer ID	Unique identifier assigned to each customer	Text
Gender	Gender of the customer (Male / Female)	Categorical
Region	Geographic region of the customer (North, South, East, West)	Categorical
Age	Age of the customer in years	Numeric
Product Name	Name of the purchased product	Categorical
Category	Product category (Electronics, Accessories, Wearables)	Categorical
Unit Price	Price per unit of the product (₹)	Numeric
Quantity	Number of units purchased in the order	Numeric
Total Price	Total order value (₹)	Numeric
Shipping Fee	Fee charged for shipping the order (₹)	Numeric
Shipping Status	Fulfilment status (Delivered, In Transit, Returned)	Categorical
Order Date	Date on which the order was placed	Date

Table 1: Dataset Attribute Description

7.1 Dataset Snapshot

- Total Records: 1,000 transactions
- Total Attributes: 12 columns
- Time Period Covered: 1 January 2023 to 3 January 2024
- Unique Customers: 292
- Products: Laptop, Smartphone, Monitor, Smartwatch, Headphones, Keyboard, Mouse
- Categories: Electronics, Accessories, Wearables
- Regions: North, South, East, West (plus records with missing region)

7.2 Nature of the Data

The dataset combines demographic attributes (Gender, Age, Region), product attributes (Product Name, Category, Unit Price), transactional attributes (Quantity, Total Price, Shipping Fee, Order Date), and an operational attribute (Shipping Status). This structure makes it suitable for a well-rounded, multi-dimensional analysis covering “who bought,” “what was bought,” “when it was bought,” and “what happened to the order.”

Chapter 8: Data Cleaning and Pre-processing

Real-world datasets are rarely analysis-ready in their raw form, and this dataset was no exception. Before any meaningful analysis could be performed, several data-quality issues had to be identified and resolved. This chapter documents each issue and the treatment applied.

8.1 Missing Value Analysis

An initial audit of the dataset revealed missing values in three columns, as summarized in Table 2.

Column	Missing Values	% of Records
Region	50	5.0%
Age	100	10.0%
Shipping Status	50	5.0%

Table 2: Summary of Missing Values Before Cleaning

8.2 Inconsistent Price Records

A cross-check of Unit Price × Quantity against the recorded Total Price revealed 20 rows (2% of the dataset) where the two values did not match, with discrepancies as large as ₹8,000 in a single order. Closer inspection showed that each product otherwise follows a fixed, standardized unit price (e.g., every Laptop is priced at ₹1,500 and every Mouse at ₹30); the mismatched rows contained randomly perturbed Unit Price values that did not correspond to this standard pricing. These were treated as data-entry noise.

8.3 Cleaning Actions Applied

Issue	Action Taken	Records Affected
Missing Region	Labelled as “Unknown” to preserve record count without fabricating a region	50
Missing Age	Imputed with the dataset median age (49 years)	100
Missing Shipping Status	Labelled as “Unknown” fulfilment status	50
Inconsistent Unit Price	Standardized to the fixed per-product price and Total Price recalculated (Unit Price × Quantity)	20
Duplicate Records	Checked — none found	0

Table 3: Data Cleaning Actions Summary

8.4 Feature Engineering

To support deeper analysis, the following derived features were engineered from the existing columns:

- Age Group: Age was bucketed into six bands (18–24, 25–34, 35–44, 45–54, 55–64, 65+) to enable clearer demographic comparisons.
- Month and Month Name: Extracted from Order Date to support monthly trend analysis.
- Quarter: Extracted from Order Date to support quarterly revenue analysis.

8.5 Post-Cleaning Summary

After cleaning, the dataset retained all 1,000 original records (no rows were dropped), with all missing values addressed and pricing inconsistencies corrected. This preserved the maximum amount of information while ensuring the integrity of downstream calculations such as total revenue and average order value.

Metric	Value
Total Revenue	₹13,46,600
Total Orders	1,000
Unique Customers	292
Total Units Sold	3,008
Average Order Value	₹1,346.60
Average Items per Order	3.01
Average Shipping Fee	₹12.42
Overall Return Rate	30.8%

Table 4: Key Performance Indicators (KPIs) Summary

Chapter 9: Exploratory Data Analysis and Results

This chapter presents the core exploratory data analysis of the cleaned e-commerce sales dataset. Each sub-section addresses a specific analytical question, supported by a chart generated in Python (Matplotlib) and followed by a written interpretation of the observed pattern and its business implication.

9.1 Revenue Distribution by Product Category

The dataset covers three product categories: Electronics, Accessories, and Wearables. Figure 1 shows the total revenue generated by each category.

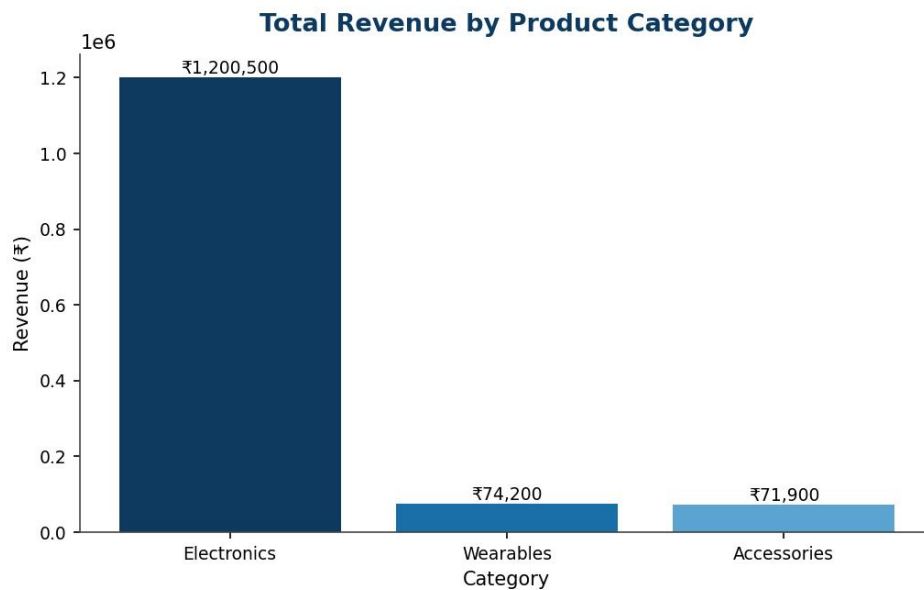


Figure 1: Total Revenue by Product Category

Electronics overwhelmingly dominates revenue generation, contributing approximately ₹12,00,479 — about 89.15% of total revenue — while Accessories and Wearables contribute only 5.34% and 5.51% respectively. This concentration is explained by the significantly higher unit prices of Electronics items (Laptops at ₹1,500 and Smartphones at ₹800) compared to Accessories (Mouse, Keyboard, Headphones, priced between ₹30 and ₹100). This finding highlights a classic Pareto-style revenue concentration, where a small number of high-value product lines account for the vast majority of sales value.

9.2 Revenue Distribution by Individual Product

Breaking revenue down further by individual product (Figure 2) provides a more granular view of which specific items drive sales.

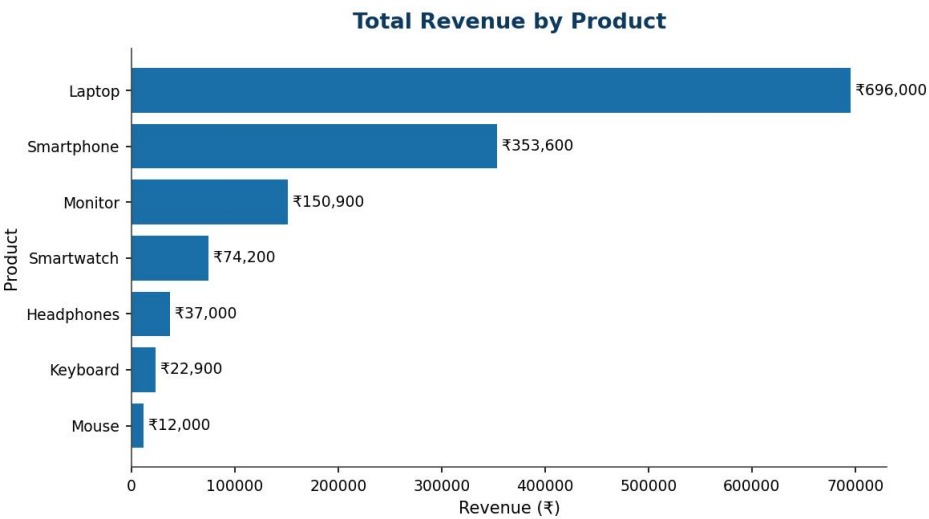


Figure 2: Total Revenue by Product

Laptops emerge as the single largest revenue contributor, followed by Smartphones and Monitors — all three being Electronics items. Despite having a comparable or even higher order count in some cases, Accessories such as the Mouse and Keyboard contribute minimally to revenue owing to their low price points. This confirms that revenue leadership in this business is driven primarily by high-ticket electronics rather than order volume alone, an important consideration for inventory and marketing budget allocation.

9.3 Order Volume by Region

Figure 3 shows the number of orders placed from each geographic region.

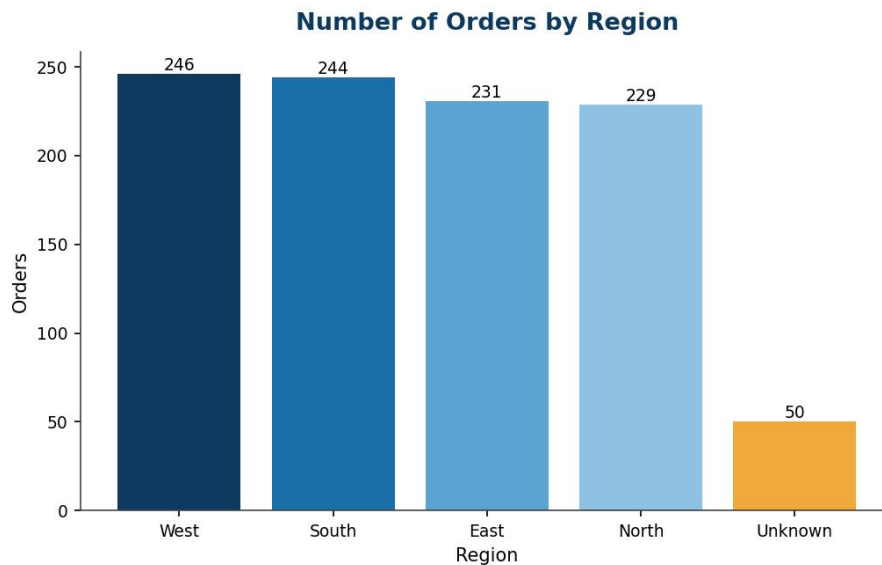


Figure 3: Number of Orders by Region

Order volume is fairly evenly distributed among the North, South, East, and West regions, each accounting for roughly a quarter of total orders, with a small proportion (5%) of orders having an unrecorded ("Unknown") region due to missing data. This even spread suggests the business has achieved a balanced geographic footprint rather than being concentrated in a single market.

9.4 Revenue Distribution by Region

Figure 4 presents the total revenue generated from each region.

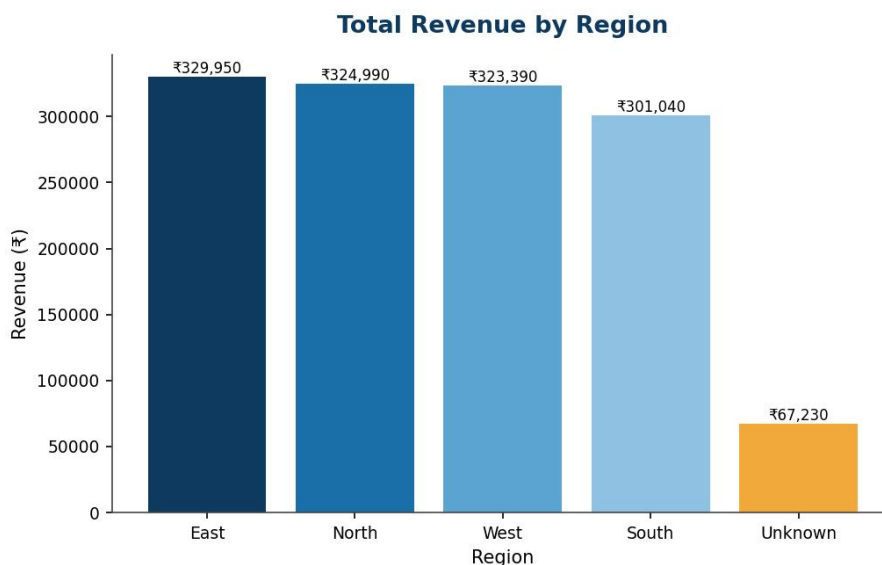


Figure 4: Total Revenue by Region

Consistent with the order-volume pattern, revenue is also fairly evenly distributed across regions — East leads marginally with 24.5% of revenue, closely followed by North (24.13%) and West (24.02%), with South slightly lower at 22.36%. The near-uniform distribution indicates that no single region is

disproportionately driving — or lagging behind — overall sales performance, suggesting consistent demand across the customer base's geography.

9.5 Monthly Revenue Trend

Understanding how revenue fluctuates over time is essential for demand forecasting and inventory planning. Figure 5 plots total monthly revenue across the twelve-month period covered by the dataset.



Figure 5: Monthly Revenue Trend (Jan 2023 - Jan 2024)

Revenue peaks in January (₹1,52,740) and November (₹1,47,150), likely reflecting New Year and pre-festive/holiday-season shopping behaviour, while March records the lowest revenue of the year (₹67,660). A secondary peak is visible in June. This seasonal pattern suggests the business should plan higher inventory levels and staffing ahead of January and November, while using quieter months like March and February for stock replenishment, maintenance, or promotional campaigns to stimulate demand.

9.6 Quarterly Revenue Distribution

Aggregating the monthly data into quarters (Figure 6) provides a smoother view of seasonal trends.

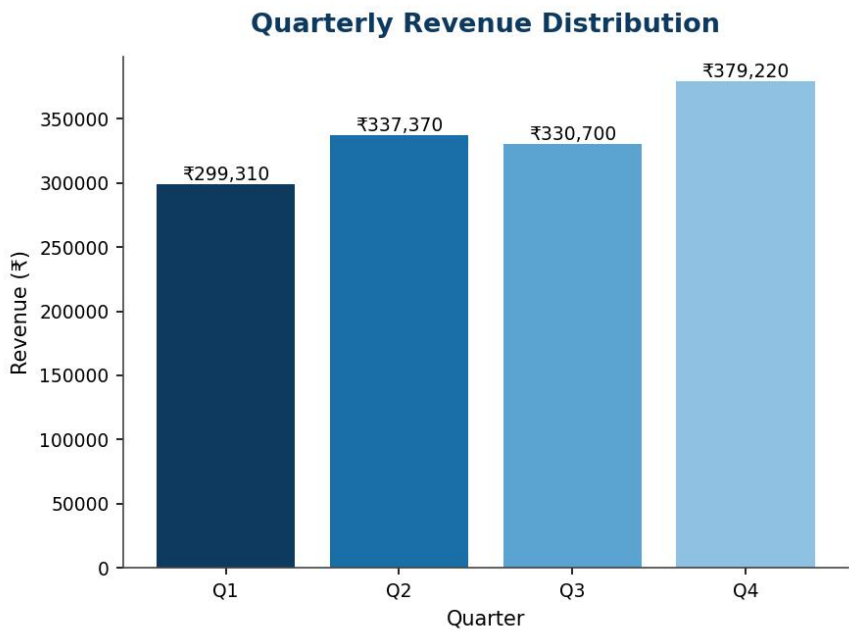


Figure 6: Quarterly Revenue Distribution

Revenue is strongest in Q4 (October–December) and Q1 (January–March, largely driven by January), while Q1 also contains the weakest month (March), producing more month-to-month volatility within that quarter. Q2 and Q3 show steadier, moderate performance. This reinforces the recommendation to align promotional and inventory strategy with the Q4-to-January demand surge.

9.7 Customer Gender Distribution

Figure 7 shows the proportion of orders placed by male versus female customers.

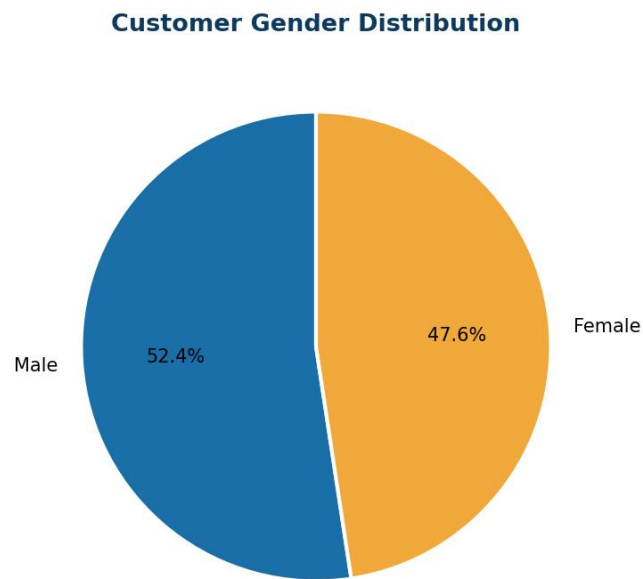


Figure 7: Customer Gender Distribution

The customer base is fairly balanced, with male customers accounting for 52.4% of orders and female customers 47.6%. In revenue terms, male customers contributed ₹7,15,590 versus ₹6,31,010 from female customers — broadly proportional to the order split. This suggests the product mix (largely electronics) appeals similarly across genders, without a strong gender skew that would necessitate dramatically different marketing strategies for each segment.

9.8 Customer Distribution by Age Group

Figure 8 breaks down the customer base into six age bands.

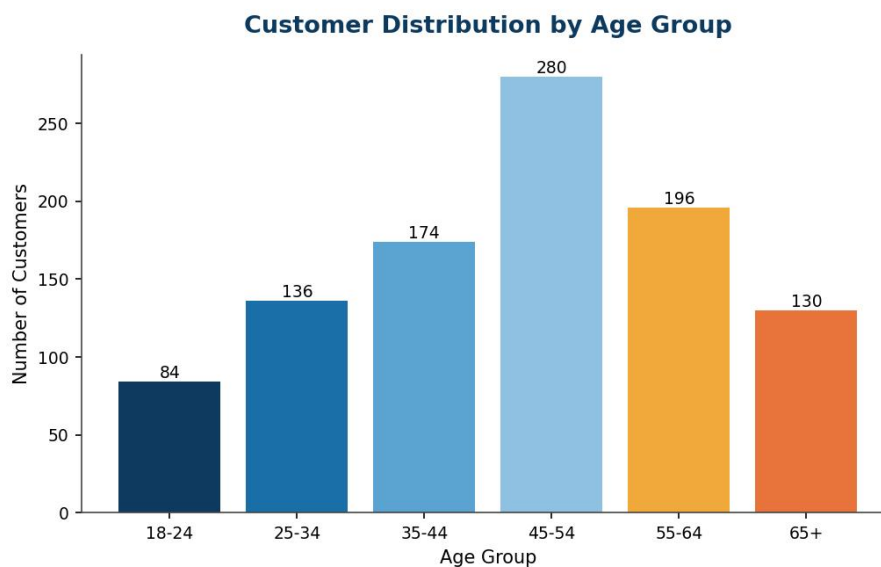


Figure 8: Customer Distribution by Age Group

Customers in the 45–54 and 55–64 age bands form the largest share of the customer base, indicating that the platform's electronics-heavy product mix resonates strongly with middle-aged and older customers rather than being skewed toward a younger, typically more tech-native demographic. This is a notable and actionable insight — marketing creatives and product recommendations could be tailored to this demographic's preferences.

9.9 Revenue Contribution by Age Group

Figure 9 shows how total revenue is distributed across the same age bands.

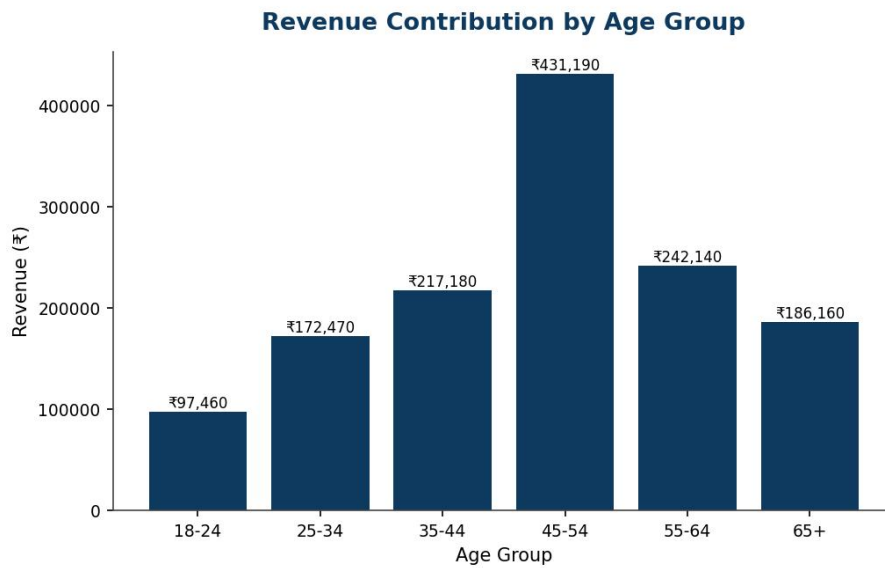


Figure 9: Revenue Contribution by Age Group

Revenue contribution closely tracks customer count by age group, with the 45–54 and 55–64 bands again leading in total spend. This alignment between customer volume and revenue share (rather than one age group over-indexing on spend) suggests fairly uniform average order values across age groups, which is explored further in the demographic cross-analysis in Section 9.16.

9.10 Order Shipping Status Distribution

Order fulfilment outcomes are a critical operational metric. Figure 10 shows the distribution of orders by shipping status.

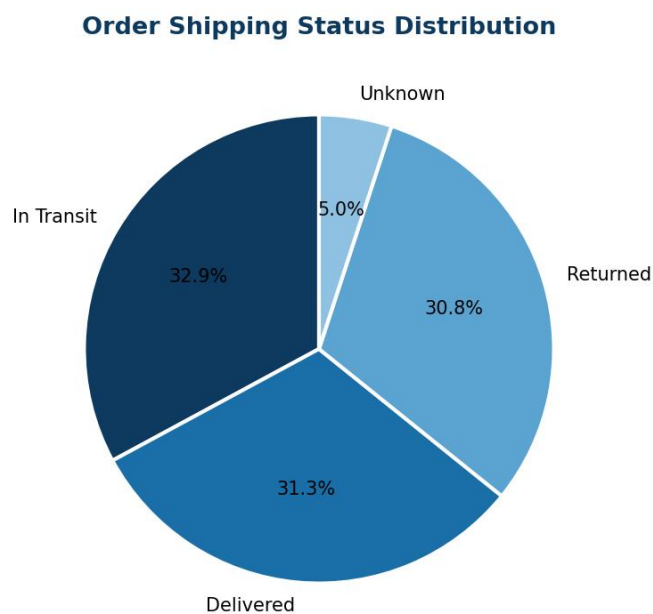


Figure 10: Order Shipping Status Distribution

Orders are split almost evenly across three statuses: In Transit (32.9%), Delivered (31.3%), and Returned (30.8%), with 5% of orders lacking a recorded status. The near-equal three-way split is notable — a 30.8% return rate is considerably higher than the 8–10% return rate typically considered healthy for electronics-heavy e-commerce businesses, and represents a significant area of concern addressed further in Chapter 10 and Chapter 11.

9.11 Average Order Value by Category

Figure 11 compares the average order value (AOV) across the three product categories.

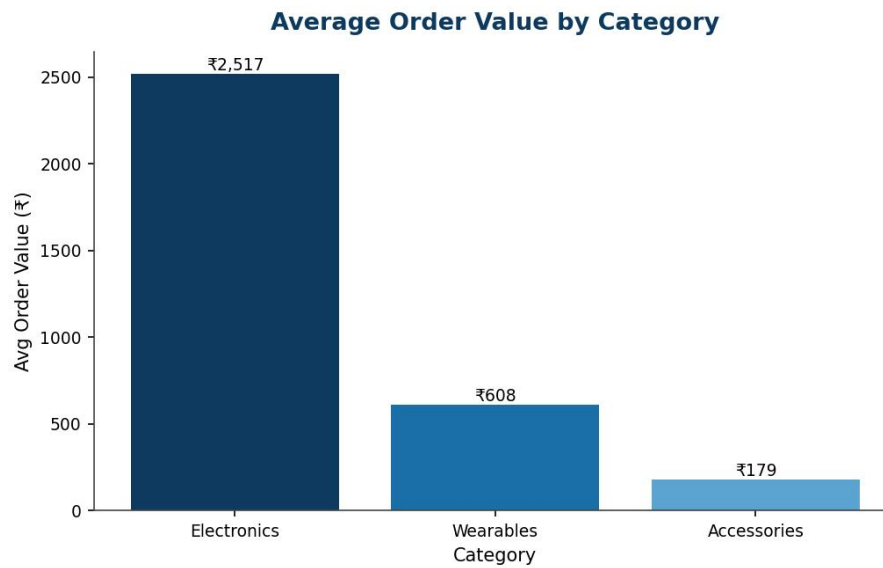


Figure 11: Average Order Value by Category

As expected given their higher unit prices, Electronics orders carry a substantially higher average order value than Accessories or Wearables. This reinforces Electronics as the category with the greatest per-transaction revenue impact, making it the natural priority for stock availability, targeted upselling, and premium customer service.

9.12 Distribution of Order Quantity

Figure 12 shows the distribution of the number of units purchased per order.



Figure 12: Distribution of Order Quantity

Order quantities are fairly evenly distributed between 1 and 5 units per order, with an average of approximately 3.01 units per order. There is no extreme skew toward single-unit purchases, suggesting

that customers frequently buy in small bulk (e.g., purchasing multiple accessories or gifting items), which has implications for packaging and bulk-discount strategy.

9.13 Correlation Analysis of Numerical Variables

To understand relationships among the numerical fields, a correlation matrix was computed for Age, Unit Price, Quantity, Total Price, and Shipping Fee (Figure 13).

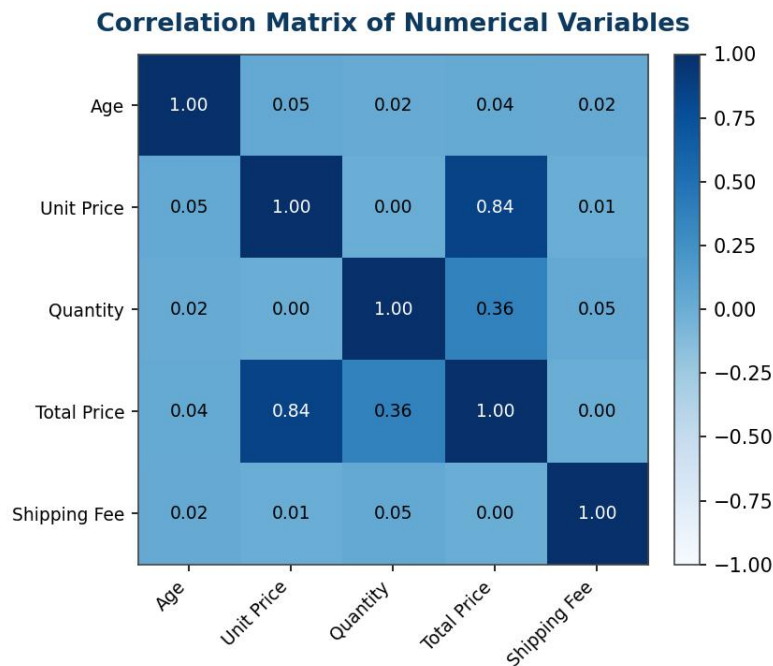


Figure 13: Correlation Matrix of Numerical Variables

As expected, Total Price shows a strong positive correlation with Unit Price, since Total Price is directly derived from it. Quantity shows a weak-to-moderate correlation with Total Price, reflecting that revenue in this dataset is driven more by which product is purchased (its price tier) than by how many units are bought. Age and Shipping Fee show negligible correlation with the other numerical variables, indicating that customer age does not meaningfully influence spending behaviour or shipping cost in this dataset.

9.14 Top Customers by Total Spend

Identifying high-value customers is important for loyalty and retention strategy. Figure 14 ranks the top 10 customers by cumulative spend.

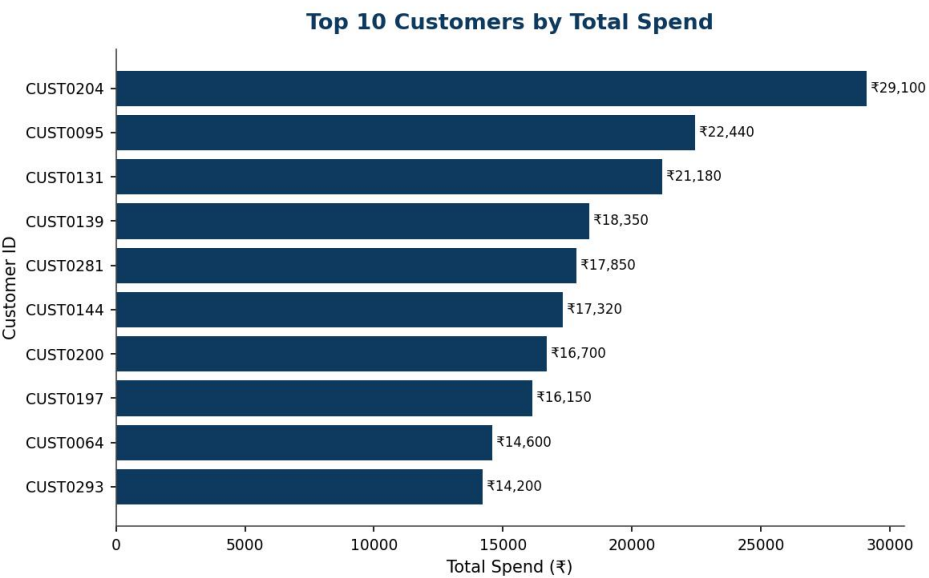


Figure 14: Top 10 Customers by Total Spend

The top customer (Customer ID CUST0204) accounts for approximately ₹29,100 in total spend — more than twenty times the average order value — indicating a highly engaged, repeat-purchase customer. Identifying and nurturing this small group of top spenders through loyalty programs or personalized offers represents a high-leverage opportunity, since retaining a small number of high-value customers is typically far more cost-effective than acquiring new ones.

9.15 Category Performance Across Regions

Figure 15 presents a heatmap of order counts by category and region, revealing whether certain product categories are more popular in specific regions.

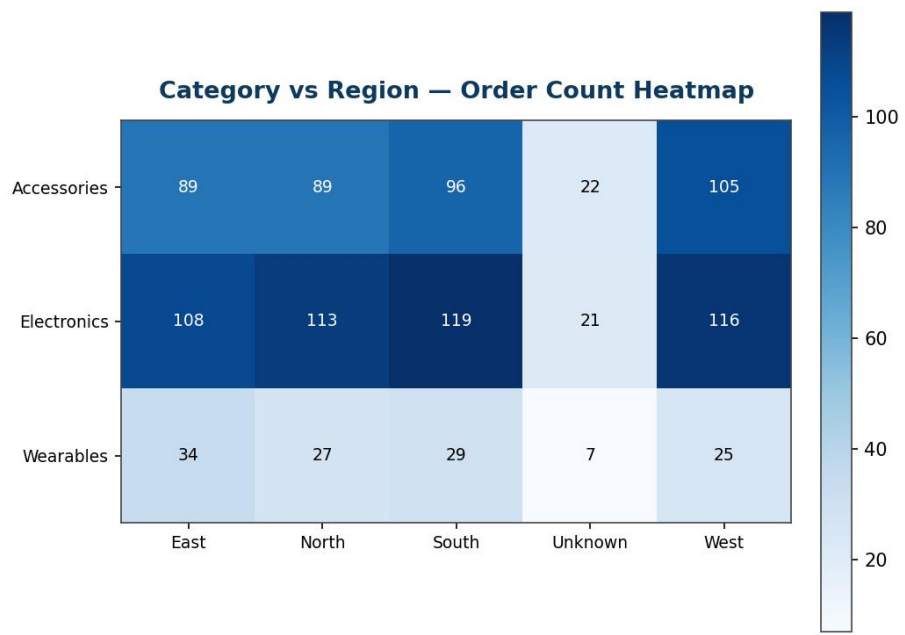


Figure 15: Category vs Region — Order Count Heatmap

Electronics orders are the most numerous across every region, consistent with its dominant overall market share, while Accessories and Wearables show more modest, fairly consistent order counts across regions. No single region shows a dramatically different category preference from the others, suggesting that a uniform, rather than region-specific, product and marketing strategy is currently appropriate.

9.16 Category Preference by Gender

Figure 16 compares product-category preferences between male and female customers.

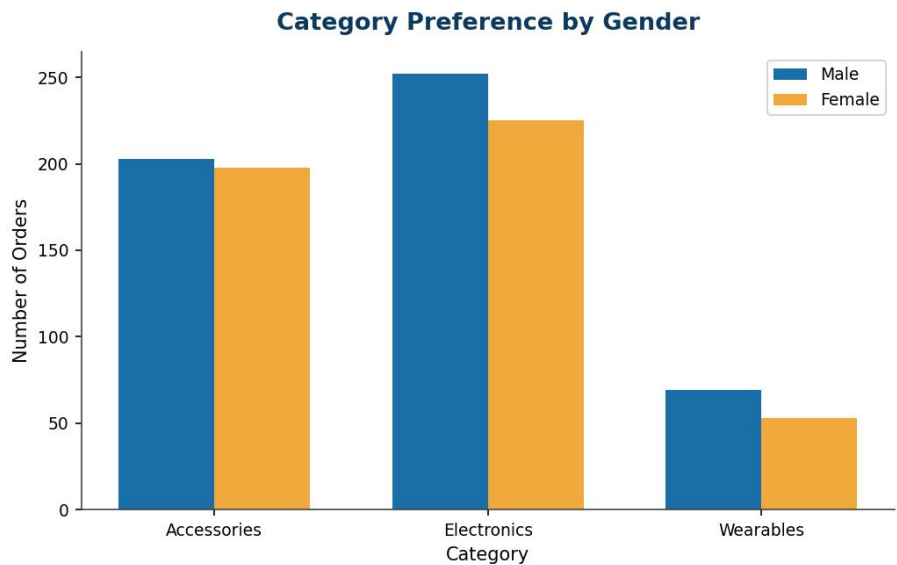


Figure 16: Category Preference by Gender

Both male and female customers show a strong preference for Electronics over Accessories and Wearables, with only modest differences between genders within each category. This supports the earlier observation (Section 9.7) that gender is not a strong differentiator of purchasing behaviour in this dataset, and that category-based (rather than gender-based) segmentation is likely to be more effective for this business.

9.17 Return Rate by Category

Given the notably high overall return rate identified in Section 9.10, Figure 17 breaks the return rate down by product category to identify which categories are most affected.

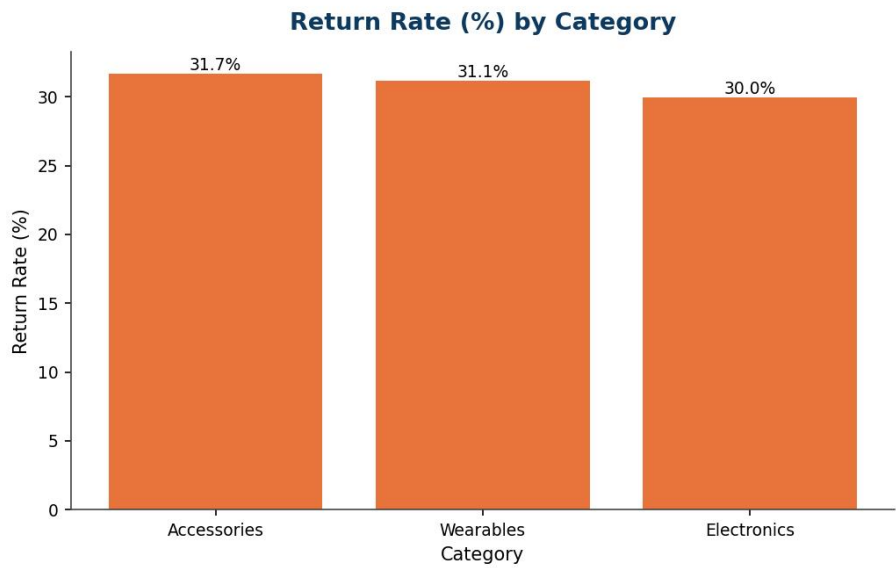


Figure 17: Return Rate (%) by Category

Return rates are elevated across all three categories, broadly in line with the overall 30.8% figure, with only minor variation between Electronics, Accessories, and Wearables. Because returns do not appear to be concentrated in a single category, the root cause is likely systemic — potentially related to product-description accuracy, packaging/damage during transit, delivery delays, or ease of the returns process — rather than being specific to any one product line. This is examined further in the recommendations presented in Chapter 11.

Chapter 10: Key Findings

The exploratory analysis conducted in Chapter 9 surfaced several important, actionable findings, summarized below:

10.1 Revenue and Product Insights

- Electronics is the dominant category, generating 89.15% of total revenue, led by Laptops as the single highest-revenue product.
- Accessories and Wearables together contribute less than 11% of revenue despite representing a meaningful share of order volume, reflecting their low unit price.
- Total revenue across all 1,000 orders stands at ₹13,46,600, with an average order value of ₹1,346.60.

10.2 Customer and Demographic Insights

- The customer base is nearly evenly split by gender (52.4% Male, 47.6% Female), with no strong gender-based skew in category preference.
- Customers aged 45–64 form the largest and highest-spending demographic segment, an important insight for age-targeted marketing.
- A small group of high-value repeat customers (e.g., the top 10 spenders) contribute disproportionately to revenue, underscoring the value of a loyalty/retention strategy.

10.3 Regional Insights

- Order volume and revenue are both fairly evenly distributed across the North, South, East, and West regions, with East marginally leading.
- No single region shows a dramatically different category preference, suggesting current strategy can remain broadly uniform across geographies.

10.4 Seasonal Insights

- January and November are peak revenue months, likely tied to New Year and pre-festive shopping behaviour.
- March is the weakest month of the year, presenting an opportunity for targeted promotions to boost off-season demand.

10.5 Operational Insights

- The overall return rate of 30.8% is significantly above the healthy benchmark range (typically 8–10%) for electronics-heavy e-commerce, representing a material drag on effective revenue.
- Elevated return rates are observed consistently across all product categories, pointing to a systemic rather than product-specific cause.
- 5% of records have missing or unrecorded shipping status, region, or age data, indicating a data-quality gap in the order-management or CRM system that should be addressed at the source.

Chapter 11: Recommendations

Based on the key findings presented in Chapter 10, the following recommendations are proposed to help the business improve revenue, customer satisfaction, and operational efficiency.

11.1 Inventory and Product Strategy

15. Prioritize stock availability and supplier reliability for Electronics — particularly Laptops and Smartphones — given their outsized contribution to revenue.
16. Bundle low-margin Accessories (Mouse, Keyboard, Headphones) with high-value Electronics purchases to increase average order value without requiring separate marketing spend.
17. Use the March sales dip to schedule stock replenishment, system maintenance, or clearance promotions rather than treating it as lost opportunity.

11.2 Marketing and Customer Engagement

18. Design targeted campaigns around the 45–64 age demographic, which currently drives the largest share of both orders and revenue.
19. Introduce a loyalty or rewards program aimed at the top-spending customer segment identified in Section 9.14, since retaining high-value repeat customers is more cost-effective than new customer acquisition.
20. Increase promotional intensity ahead of the identified peak months (January and November) to fully capitalize on naturally higher demand.

11.3 Reducing Return Rates

21. Conduct a root-cause investigation into the 30.8% return rate, examining product-description accuracy, packaging quality, and delivery-partner performance, since returns are elevated uniformly across categories rather than isolated to one product line.
22. Improve product imagery, specifications, and size/compatibility information at the point of sale to reduce expectation-mismatch returns.
23. Introduce a post-delivery feedback or quality-check step for high-value Electronics orders, where the cost of a return is greatest.

11.4 Data Quality and Systems

24. Enforce mandatory field capture for Region, Age, and Shipping Status at the point of order entry to eliminate the current 5–10% missing-data gap in these fields.
25. Implement automated validation rules (e.g., Total Price = Unit Price × Quantity) at the database or application layer to prevent the pricing inconsistencies identified during data cleaning from recurring.
26. Establish a recurring (e.g., monthly) sales-analytics reporting cycle using the methodology demonstrated in this project, so that findings can inform decisions on an ongoing basis rather than as a one-time exercise.

Chapter 12: Conclusion

This project set out to analyze an e-commerce sales dataset comprising 1,000 transactions in order to extract meaningful business insights using core data-mining and data-visualization techniques. Through a structured process of data cleaning, exploratory analysis, and visualization using Python, pandas, and Matplotlib, the study successfully uncovered actionable patterns across product performance, customer demographics, regional distribution, seasonality, and order fulfilment.

The analysis confirmed that Electronics — particularly Laptops and Smartphones — are the primary revenue drivers for this business, contributing nearly 90% of total revenue despite representing less than half of the product catalogue. Customer demographics revealed a balanced gender split but a pronounced concentration of spending among customers aged 45 to 64. Seasonal analysis identified January and November as peak months, while March emerged as the weakest period, offering clear guidance for inventory and promotional planning.

Perhaps the most operationally significant finding was the elevated overall return rate of 30.8%, which is considerably higher than industry benchmarks and consistent across all product categories — suggesting a systemic issue worth deeper investigation. Additionally, the data-cleaning phase itself proved valuable, surfacing data-quality gaps (missing region, age, and shipping-status fields, along with inconsistent pricing records) that point to opportunities for improving data capture at the source.

Overall, this project demonstrates how even a modestly sized transactional dataset, when subjected to careful cleaning and thoughtful exploratory analysis, can yield rich, decision-relevant insights. The recommendations put forward in Chapter 11 — spanning inventory strategy, targeted marketing, return-rate reduction, and data-quality improvement — provide a concrete, evidence-based roadmap that the business could act upon to improve both profitability and customer experience.

Chapter 13: Future Scope

While this project provides a comprehensive descriptive and exploratory analysis, several avenues exist for extending this work in the future:

- **Predictive Modelling:** Building machine-learning models (e.g., regression or time-series forecasting) to predict future monthly sales and support proactive inventory planning.
- **Customer Segmentation:** Applying clustering algorithms such as K-Means, potentially combined with RFM (Recency, Frequency, Monetary) analysis, to identify distinct customer personas beyond simple demographic groupings.
- **Return Prediction:** Developing a classification model to flag orders at high risk of being returned at the time of purchase, enabling proactive intervention.
- **Market Basket Analysis:** Applying association-rule mining to uncover which products are frequently purchased together, to inform cross-selling and bundling strategies.
- **Larger and Richer Datasets:** Extending the analysis to a larger transaction volume with additional attributes (e.g., marketing channel, discount codes, customer reviews) for deeper insight.
- **Real-Time Dashboards:** Converting the static analysis in this report into an interactive, real-time business-intelligence dashboard (e.g., using Power BI, Tableau, or a Python-based dashboard framework) for continuous monitoring.

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Appendix: Python Source Code

The following Python code (using pandas and Matplotlib) was used to clean the dataset, compute key metrics, and generate the visualizations presented in this report. It is reproduced here for reference and reproducibility.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. Load the raw dataset
df = pd.read_csv('E-commerce_sales_data.csv')

# 2. Fix inconsistent Unit Price / Total Price records
clean_price_map = {'Mouse':30, 'Keyboard':50, 'Headphones':100,
                  'Smartwatch':200, 'Monitor':300,
                  'Smartphone':800, 'Laptop':1500}
df['Unit Price'] = df['Product Name'].map(clean_price_map)
df['Total Price'] = df['Unit Price'] * df['Quantity']

# 3. Handle missing values
df['Region'] = df['Region'].fillna('Unknown')
df['Shipping Status'] = df['Shipping Status'].fillna('Unknown')
df['Age'] = df['Age'].fillna(df['Age'].median())

# 4. Feature engineering
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Month'] = df['Order Date'].dt.month_name()
df['Quarter'] = df['Order Date'].dt.quarter

def age_group(a):
    if a < 25: return '18-24'
    elif a < 35: return '25-34'
    elif a < 45: return '35-44'
    elif a < 55: return '45-54'
    elif a < 65: return '55-64'
    else: return '65+'
df['Age Group'] = df['Age'].apply(age_group)

# 5. Key metrics
total_revenue = df['Total Price'].sum()
avg_order_value = df['Total Price'].mean()
return_rate = (df['Shipping Status']=='Returned').mean()*100

# 6. Example visualization: Revenue by Category
cat_rev = df.groupby('Category')['Total Price'].sum()
cat_rev.sort_values(ascending=False).plot(kind='bar', color='#1B6FA8')
plt.title('Total Revenue by Product Category')
plt.ylabel('Revenue (Rs.)')
plt.tight_layout()
plt.savefig('revenue_by_category.png', dpi=150)
```